# CRYSTAL CHENG

cjcheng@berkeley.edu | linkedin.com/in/chengjcrystal | github.com/chengjcrystal | chengjcrystal.vercel.app

## EDUCATION

**University of California, Berkeley,** Berkeley, CA — Expected Dec 2026

***Bachelor of Arts in Computer Science & Applied Mathematics | GPA: 3.71***

- Coursework: Artificial Intelligence, Machine Learning, Data Structures, Statistical Modeling, Linear Algebra, Probability Theory
- Involvements: Society of Women Engineers (Treasurer), Web Development at Berkeley (Developer)

## WORK EXPERIENCE

**Hyve Solutions | Software Engineer Intern** — June 2026 - Aug 2026

- Fixed an Azure LLM token-limit failure blocking concurrent users by replacing full-catalog retrieval with Hugging Face embeddings and Neo4j vector search, capping retrieval to a 150-item top-K with a keyword fallback for safe rollout.
- Built a 5-stage hybrid retrieval pipeline (phrase match, keyword, fuzzy prefilter, vector search, LLM re-rank) that ranks test cases from natural-language queries, later adopted as the shared retrieval layer behind two production features.
- Designed and delivered the UI/UX for an AI test-planning tool on a 5-person team: produced 100% of new-feature mockups, merged 11 PRs as primary author, and shipped 6 workflows from design through deployment.

**TrackFly | Software Engineer Intern** — Sep 2025 - Dec 2025

- Reduced cross-team integration mismatches on a 15-person team by designing REST API contracts and MySQL schemas that defined a single source of truth for feature data across services.
- Cut per-feature build time by shipping 15+ reusable React/TypeScript components that standardized UI patterns across 10 weeks.

**Barobo | Software Engineer Intern** — May 2025 - Aug 2025

- Eliminated unauthorized video sharing across all user traffic by engineering single-use AWS CloudFront signed URLs with per-user access control and configurable TTL expiration.
- Scaled to 10,000+ monthly authorized video requests by tracking user access metadata in AWS DynamoDB, enabling reliable per-request authorization without adding infrastructure overhead.
- Cut environment provisioning from 2 hours to 15 minutes by rewriting deployment as reusable Terraform and AWS CLI scripts, making infrastructure reproducible across environments.

**Opal Mentorship App | Machine Learning Intern** — May 2024 - May 2025

- Raised mentor-mentee match accuracy 12% across cohorts by calibrating 6 scoring weights over a 2-round data-backed process.
- Built a reproducible ML evaluation pipeline in AWS Lambda and DynamoDB that scores compatibility across 4 behavioral dimensions, giving the team versioned, auditable weighting decisions at cohort scale.

## PROJECTS

**RepoRank Github Search Engine |** *Python, FastAPI, SQLAlchemy* — *link*

- Served BM25/BM25F search over 157k GitHub repos on a from-scratch field-aware inverted index: p95 23.8ms uncached, sub-millisecond median cached (94.3% hit rate), single-process throughput saturating at ~100 QPS (GIL-bound).
- Built an offline ranking-eval gate (nDCG/MRR/P@k with bootstrap confidence intervals over hand-labeled queries) wired into CI against a frozen 157k index, failing the build when nDCG regresses past a set margin.
- Crawled and indexed 157k repos with a resumable rate-limited GitHub crawler, rebuilding a field-aware inverted index snapshot

**FreshCheck Fruit Freshness Classifier |** *PyTorch, ONNX, React, Vercel* — *link*

- Ran sub-10ms fruit-freshness inference fully in-browser with zero backend by training a 24K-parameter CNN in PyTorch and exporting to ONNX (WebAssembly), keeping all images on-device.
- Caught and fixed train/test leakage from duplicate source images by re-splitting on content and perceptual hashes into a leak-free held-out set, reporting honest 97.7% test accuracy (43/44) with a 95% confidence interval.

**MBTI Guesser |** *Hugging Face Transformers BART-MNLI, DeepFace, OpenCV, Python* — *link*

- Designed a 13-feature multimodal sensing pipeline fusing text, visual, and behavioral signals for zero-shot personality classification across 16 MBTI types without labeled training data or a fine-tuning step.
- Combined NLI-based inference (BART-MNLI), computer vision (DeepFace/OpenCV), and weighted late fusion with dynamic signal reweighting to adapt to missing modalities at inference time, maintaining prediction quality under partial input availability.
- Mitigated short-input ENTJ bias by redesigning NLI hypothesis templates and adding per-dimension confidence thresholds to filter low-confidence false positives, reducing type collapse across skewed input distributions.

## SKILLS

- ML & Data: PyTorch, TensorFlow, HuggingFace Transformers, ONNX, RAG, Sklearn, NumPy, Pandas, OpenCV
- Cloud, DevOps, & Tools: AWS (S3, CloudFront, DynamoDB, Lambda), Docker, Jenkins, Terraform, Git/Github, Linux, Postman
- Languages: Python, C/C++, Javascript, TypeScript, SQL, Bash, HTML/CSS (Tailwind)
- Frameworks & Frontend: React, Next.js, Node.js, REST APIs